

Progress Report

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Project Details

Title	Smart Gesture-Based Communication and Health Monitoring Glove Using ESP32
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1 Problem Statement

Existing assistive technologies are often expensive, complex, or limited to a single function. There is a need for a low-cost, portable system that integrates gesture-based communication, health monitoring, and emergency alerting in a single wearable device.

2 Objectives

- Develop gesture-based communication using flex sensors.
- Convert gestures into predefined messages.
- Monitor heart rate and body temperature in real time.
- Establish emergency SOS alert functionality.
- Design a low-cost, wearable prototype using ESP32.

3 System Overview

The system consists of two units:

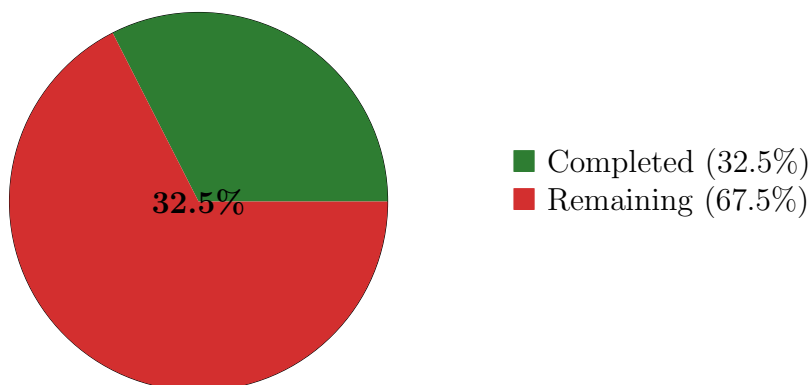
- **Glove Unit:** ESP32 with five flex sensors, pulse sensor, and DS18B20 temperature sensor.
- **Central Unit:** ESP32 with LCD, DFPlayer module, GSM module, and buzzer (to be implemented in later phases).

4 Overall Project Progress

Table 1: Phase-Wise Completion Status

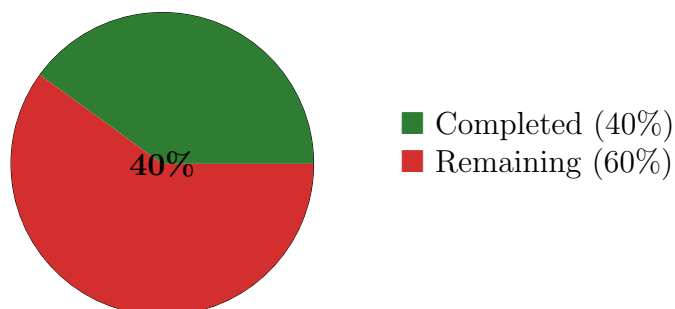
Phase	Description	Status
Phase 1	Literature Study & System Design	100%
Phase 2	Sensor Interfacing	100%
Phase 3	Gesture Detection Algorithm	100%
Phase 4	Bluetooth Communication	0%
Phase 5	Central Unit Integration	0%
Phase 6	Testing & Validation	0%
Overall Completion		32.5%

Overall Project Completion Pie Chart

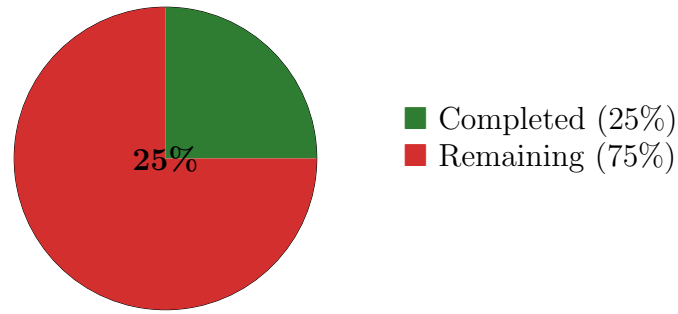


5 Module-Wise Completion Status

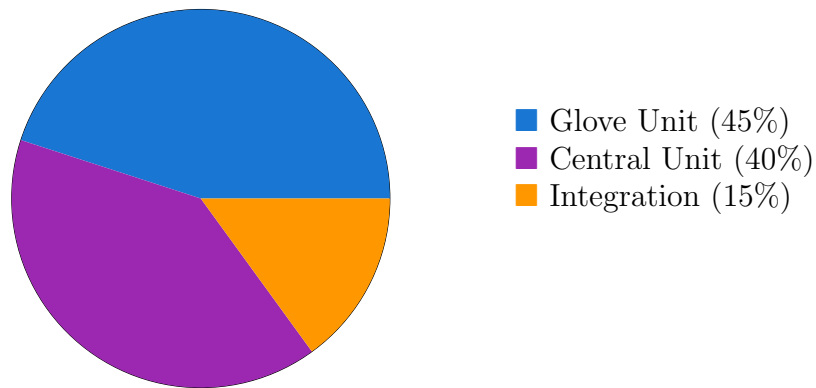
5.1 Glove Unit Completion



5.2 Central Unit Completion



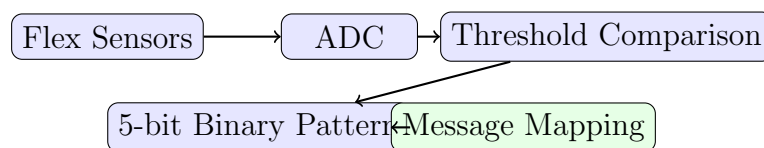
6 Work Distribution by Module



7 Gesture-to-Message Mapping (Completed)

Gesture Pattern	Message
00000	Normal / OK
11111	SOS
01111	Yes
10111	No
00111	Need Water
11110	Washroom
10011	Need Food
00011	Call / Attention

8 Gesture Detection Algorithm Workflow



9 Challenges

- Flex sensor readings vary per finger, requiring individual threshold tuning.
- Analog signal noise necessitates filtering or repeated sampling.
- Pulse sensor requires stable contact for reliable readings.

10 Conclusion

The project has successfully completed 32.5% of the planned work. All objectives for Phases 1–3 have been met:

- System architecture and component selection finalized.
- All sensors (flex, pulse, temperature) successfully interfaced.
- Gesture detection algorithm developed with 8 predefined messages.

The glove unit is 40% complete, and the central unit is 25% complete. The remaining work includes Bluetooth communication, central unit integration, and testing.

11 Next Phase (Phase 4)

- Implement Bluetooth communication between glove and central ESP32 units.
- Transmit gesture messages and health data wirelessly.
- Complete remaining glove unit and central unit integration.

12 Project Development Progress

This section details the current development status of both the glove unit and the central unit, along with identified challenges and planned additions.

12.1 Current Progress Status

Table 2: Current Progress Status

Module		Status	Remarks
Flex Sensors		Partial	Working but unstable
Temperature Sensor (DS18B20)		Done	Calibrated
OLED Display		Done	UI working
ESP-NOW Communication		Done	Communication working
Central LCD		Done	Receiving data

12.2 Current Problems and Solutions

Table 3: Current Problems and Solutions

Problem	Cause	Solution
Unknown Gesture	Threshold issue	Adjust threshold to 75–80
Wrong Detection	Noise	Add debounce logic
Index Finger Unstable	Small sensor range	Recalibrate flex sensor
Random Bending	Noise in readings	Implement filtering

12.3 Glove Unit – Component Status

Table 4: Glove Unit Component Status

Component	Purpose	Status
ESP32	Microcontroller	Done
Flex Sensors (5)	Gesture input	Partial
DS18B20	Body temperature monitoring	Done
OLED Display	Local feedback display	Done
ESP-NOW	Wireless communication	Done
MAX30102	Pulse / SpO ₂ monitoring	To Add

12.4 Central Unit – Component Status

Table 5: Central Unit Component Status

Component	Purpose	Status
ESP32	Microcontroller	Done
LCD Display	Message and health data output	Done
DFPlayer	Audio playback	To Add
Speaker	Audio output	To Add
SIM800L	GSM / SMS emergency alerts	To Add
Buzzer	Audible alerts	To Add
LED	Visual indicators	To Add

Table 6: Power System Configuration

Unit	Component	Purpose
Glove Unit	Li-ion Battery	Power supply
Glove Unit	TP4056	Battery charging module
Central Unit	18650 Battery	Power supply
Central Unit	MT3608	Boost converter for 5V output

12.5 Power System

12.6 Summary of Development Progress

- **Fully Implemented:** ESP32 controllers, DS18B20 temperature sensor, OLED display, ESP-NOW communication, Central LCD.
- **Partially Implemented:** Flex sensors (working but unstable).
- **To Be Added:** MAX30102 (pulse/SpO₂), DFPlayer, speaker, SIM800L GSM module, buzzer, LED indicators.
- **Power System:** Designed for both units using Li-ion and 18650 batteries with TP4056 charging and MT3608 boost converter.

References

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6. SIMCom, “SIM800L GSM Module Specification,” 2020.
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